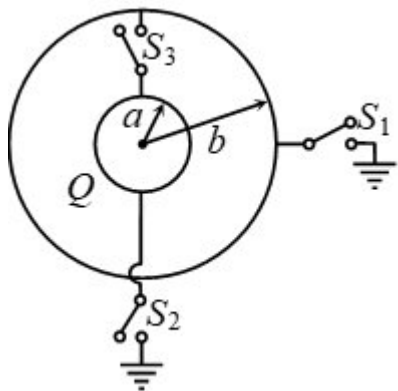


Q1

The figure shows a conducting sphere 'A' of radius ' a ' which is surrounded by a neutral conducting spherical shell 'B' of radius ' b ' ($> a$). Initially switches S_1 , S_2 and S_3 are open and sphere 'A' carries a charge Q . First the switch ' S_1 ' is closed to connect the shell 'B' with the ground and then opened. Now the switch ' S_2 ' is closed so that the sphere 'A' is grounded and then S_2 is opened. Finally, the switch ' S_3 ' is closed to connect the spheres together. If the heat (in mJ) which is produced after closing the switch S_3 is x , then find $\frac{x}{9}$. [Consider $b = 4$ cm, $a = 2$ cm and $Q = 0.8\mu\text{C}$]

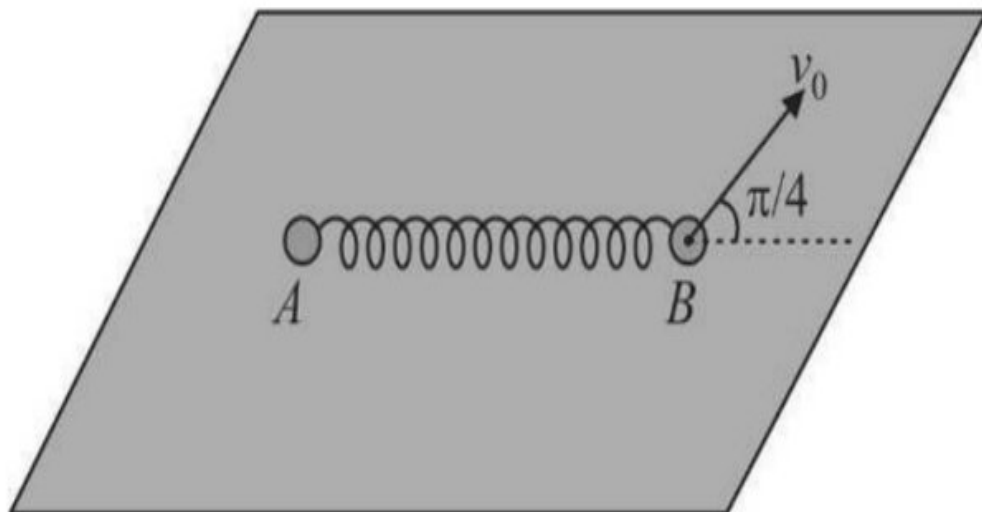


Q2

If the sound heard by observer, whose equation is given as $y = 8 \sin 10\pi t \cos 200\pi t$ at $x = 0$, then number of beat frequency heard by observer is $2k$. Then the value of k is

Q3

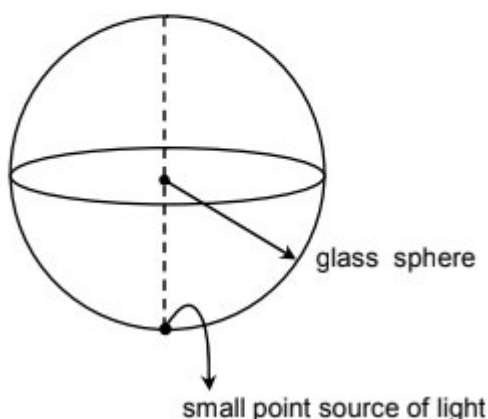
A dumbbell consists of two balls A and B each of mass $m = 1$ kg and connected by a spring. The whole system is placed on a smooth horizontal surface as shown in the figure. Initially the spring is at its natural length, the ball B is imparted a velocity $v_0 = \frac{8}{\sqrt{7}}$ m/s in the direction shown. The spring constant of the spring is adjusted so that the length of the spring at maximum elongation is twice that of the natural length of the spring. Find the maximum potential energy stored (in Joule) in the spring during the motion.



Q4

A glass sphere ($\mu = \frac{5}{3}$) with an inbuilt point source of light P is placed on table. The source P is just inside the surface of glass and lies on the lower end of vertical diameter. The apparatus is placed in a dark room. The surface area of the glowing portion of the glass sphere as observed by the eye is found to be $\left(\frac{\alpha}{\beta}\right)^2 \pi R^2$. Here α and β are integers.

Find minimum value of $\alpha - \beta$.



Q5

An isolated and charged spherical soap bubble has a radius ' r ' and the pressure inside is atmospheric. If ' T ' is the surface tension of soap solution, then charge on drop is $N\pi r\sqrt{2rT\epsilon_0}$, then find the value of N .

Q6

One mole of an ideal monatomic gas is kept in a rigid vessel whose outer surface is maintained at a constant temperature 100°C . The thickness and surface area of the vessel are 5 mm and 400 cm^2 respectively. Initial temperature of the gas in the vessel is 16°C . The conductivity of the material of the walls is $0.01\text{ W m}^{-1}\text{ K}^{-1}$. A sound signal move with a velocity $320 + n$ in the gas contained in the vessel after 250 seconds if initially its speed in the gas is 290 m/s . Find the value of n . (nearest integer)

Given $e^{1.6} = 4.94$, gas constant $R = \frac{25}{3}\text{ J mol}^{-1}\text{ K}^{-1}$. (Neglect the heat capacity of the vessel.)

Q7

Imagine an atom made up of a proton and a hypothetical particle of double the mass of the electron but having the same charge as that of the electron. Apply the Bohr atom model and consider all possible transitions of this hypothetical particle to the first excited level. The longest wavelength photon that will be emitted has wavelength λ , (given in terms of Rydberg constant R for hydrogen atom) equal to

Questions with Answer Keys

MathonGo

(1) $\frac{9}{5R}$

(2) $\frac{36}{5R}$

(3) $\frac{18}{5R}$

(4) $\frac{4}{R}$

Q8

At a point on the screen in YDSE experiment 3rd maxima is observed at $t = 0$. Now screen is slowly moved with constant speed away from the slits in such a way that the centre of slits and centre of screen lie on the same, line always and at $t = 1\text{sec}$ the intensity at that point is observed $(3/4)\text{th}$ of maximum intensity in between 2nd and 3rd maxima. The speed of screen may be ($D = \text{separation between the screen and slits}$ $d = \text{separation between the slits}$, $d \ll D$, $\lambda = 5000\text{\AA}$)

(1) $\frac{7D}{13}$

(2) $\frac{13D}{5}$

(3) $\frac{D}{17}$

(4) None of these

Q9

There is some change in length when a 33000 N tensile force is applied on a steel rod of the area of cross-section 10^{-3} m^2 . The change of temperature required to produce the same elongation, if the steel rod is heated, is (The modulus of elasticity is $3 \times 10^{11}\text{ N m}^{-2}$ and the coefficient of linear expansion of steel is $1.1 \times 10^{-5}\text{ }^\circ\text{C}^{-1}$)

(1) 20°C

(2) 15°C

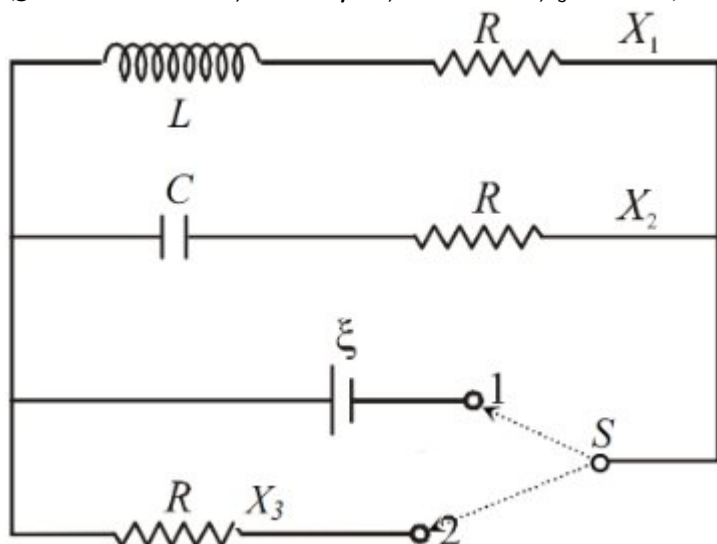
(3) 10°C

(4) 0°C

Q10

Figure shows a circuit containing three resistor X_1 , X_2 & X_3 having resistance R each, an inductor, capacitor and an emf source having inductance 'L', capacitance C & emf ξ respectively (Given $R = \sqrt{\frac{L}{C}}$). The switch is first connected to position-1. When charge on capacitor becomes half of it maximum possible value then switch S is connected to position-2. Current in resistance X_3 just after shifting the switch from position-1 to position-2, is

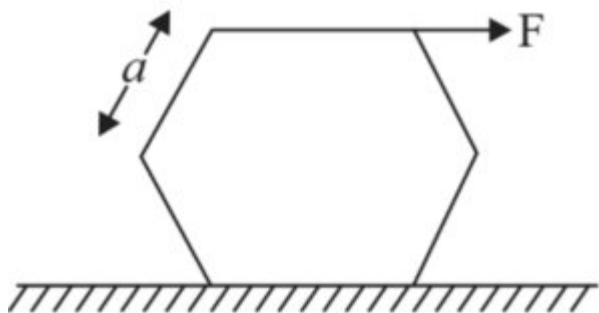
Questions with Answer Keys

(given $L = 5\text{mH}$, $C = 2\mu\text{F}$, $R = 10\Omega$, $\xi = 5\text{ V}$)

- (1) 1
 (2) 1.5
 (3) 2.5
 (4) None of these

Q11

When force F acts on side of hexagonal body for what range of coefficient of friction body will topple before slipping?



- (1) $\mu > 0.29$
 (2) $\mu < 0.29$
 (3) $\mu > 0.21$
 (4) $\mu < 0.21$

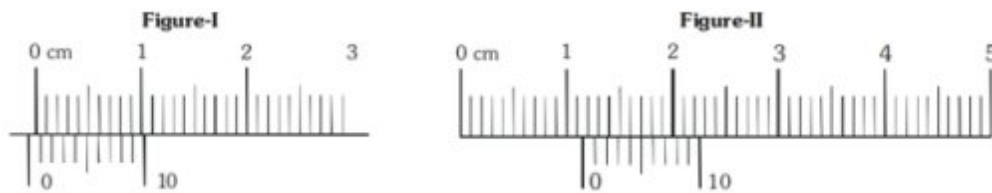
Q12

In ordinary Vernier calipers, 10^{th} division of the Vernier scale coincides with 9^{th} division of the main scale. In a specially designed Vernier calipers the Vernier scale is so constructed that 10^{th} division on it coincides with 11^{th}

Questions with Answer Keys

MathonGo

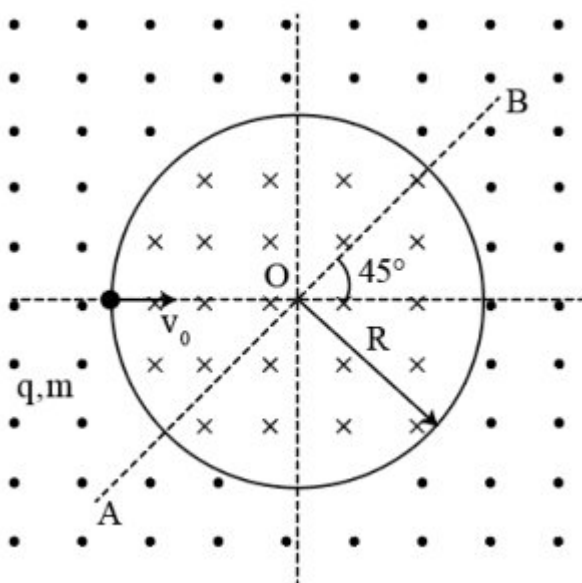
division on the main scale. Each division on the main scale equals to 1 mm. The calipers have a zero error as shown in the figure-I. When the Vernier caliper is used to measure a length, the concerned portion of its scale is shown in figure-II.



- (1) Zero error in the calipers has magnitude 0.7 mm.
- (2) The length being measured is 1.08 cm.
- (3) The length being measured is 1.22 cm.
- (4) Though the given Vernier scale does not follow the principle of Vernier scale, yet can be used satisfactorily.

Q13

A particle having mass 0.5kg and charge 0.2C is projected with a speed $V_0 = 10$ m/s from the periphery of a circle of radius $R = 5$ m towards the centre as shown in the figure such that region inside the circle carries a uniform magnetic field of strength $B_0 = 5$ T into the plane and in the outside region has a uniform magnetic field of the same strength coming out of the plane then, (Assume the center O to be the origin of the co-ordinate axis)



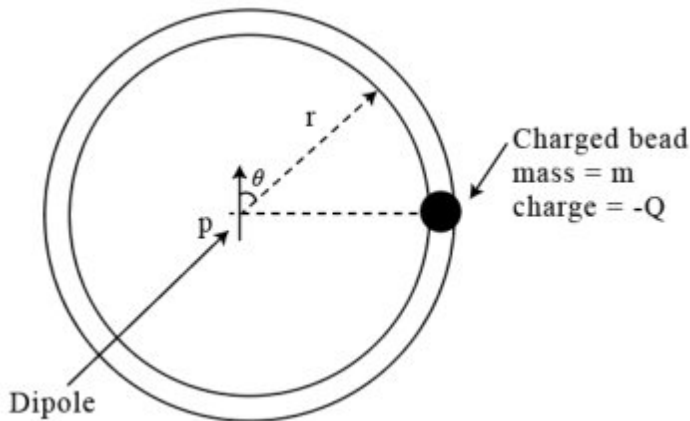
- (1) the motion of the charged particle is periodic
- (2) the particle crosses line AB after nearly 5.1 s of its start from the periphery
- (3) the force on the charged particle due to uniform magnetic field is 20 N
- (4) the co-ordinates of the center of the circle at $t = 2$ s is (5, 5)

Q14

Questions with Answer Keys

MathonGo

A small charged bead can slide on a circular frictionless, insulating wire frame. A point like dipole is fixed at the centre of circle, dipole moment is $\vec{p}$. Initially the bead is on the plane of symmetry of the dipole. Bead is released from rest. Ignore the effect of gravity. Mark the correct options



- (1) Magnitude of velocity of bead as function of its angular position is $\sqrt{\frac{Qp \cos \theta}{2\pi\epsilon_0 m r^2}}$
- (2) Normal force exerted by the string on bead is zero at all points
- (3) If the wire frame were not present bead executes circular motion and returns to initial point after tracing a complete circle.
- (4) Bead would move along a circular path until it reached the opposite its starting position and then executes periodic motion

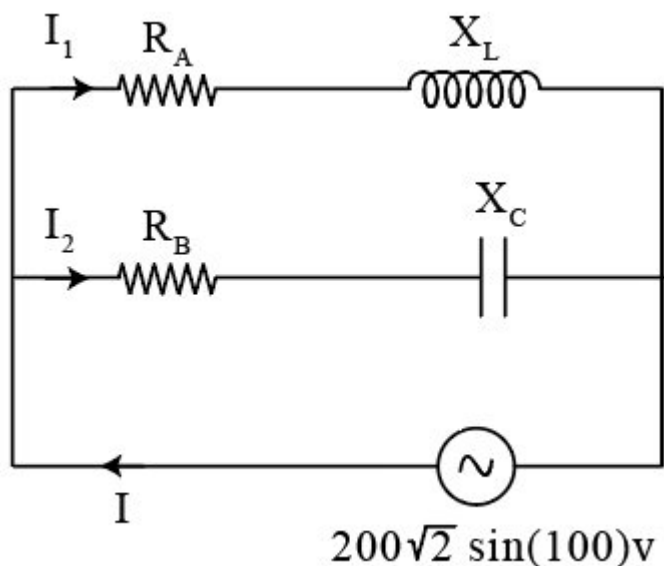
Q15

A total charge q passes through a coil of resistance R . Let H be the amount of heat generated in the coil. Choose the correct statement (s).

- (1) $H = \frac{2q^2 R}{3t_0}$ if current decreases uniformly to zero during a time interval t_0
- (2) $H = \frac{4q^2 R}{3t_0}$ if current decreases uniformly to zero during a time interval t_0
- (3) $H = \frac{q^2 R}{t_0}$ if the current decreases smoothly down to zero halving its values every t_0 seconds
- (4) $H = \frac{q^2 R \ln(2)}{2t_0}$ if the current decreases smoothly down to zero halving its value every t_0 seconds

Q16

Figure shows a two branched parallel circuit with $R_A = 10\Omega$, $L = \frac{\sqrt{3}}{10}$ H, $R_B = 20\Omega$ and $C = \frac{\sqrt{3}}{2}$ mH. Current in $L - R_A$ is I_1 and in $C - R_B$ is I_2 and main current is I



- (1) Phase difference between I_1 and I_2 is 90°
- (2) At some instant current in $L - R_A$ is 10 A. At the same instant current in $C - R_B$ branch will be $5\sqrt{3}$ A
- (3) At some instant I_1 is $10\sqrt{2}$ A then at this instant I will be $10\sqrt{2}$ A
- (4) Power dissipated in the circuit is 2121.3 W

Q17

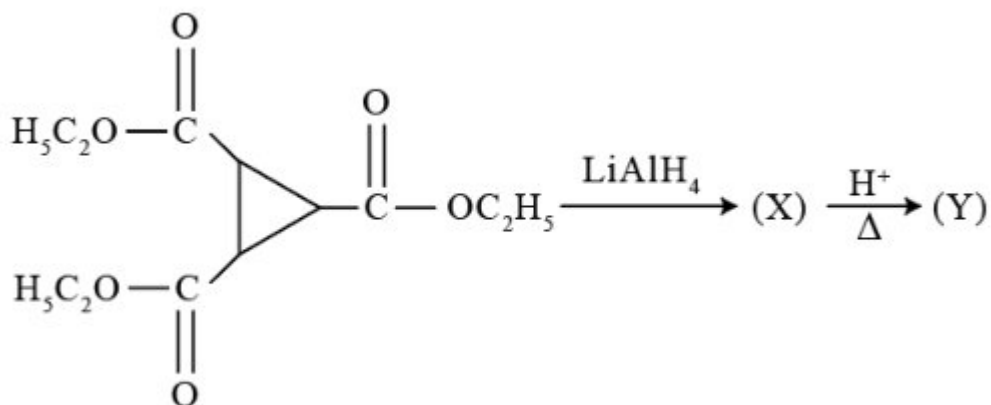
A radio active nuclei A decays in B with decay constant λ . The nucleus B may further decay in stable nuclei C or D with decay constant λ and 2λ respectively. Initially at $t = 0$, the number of nuclei of A is N_0 and number of all other nuclei is zero. If N_1, N_2, N_3 and N_4 , are number of nuclei of A, B, C and D respectively at any time t . Then select correct relations.

- (1) $N_1 = N_0 e^{-\lambda t}$
- (2) $N_2 = \frac{N_0}{2} (e^{-\lambda t} - e^{-3\lambda t})$
- (3) $N_3 = \frac{1}{2} (N_0 - N_1 - N_2)$
- (4) $N_4 = \frac{2}{3} (N_0 - N_1 - N_2)$

Q18

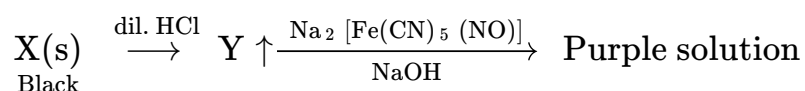
Consider the following reaction, $[\text{Ag}(\text{CN})_2]^- \rightleftharpoons \text{Ag}^+ + 2\text{CN}^-$, the K_C at 25°C is 4×10^{-19} . The concentration of Ag^+ is $x \times 10^{-18}$ M in solution which was originally 0.1M in KCN^- and 0.03M in AgNO_3 . The value of x is

Q19



Find the number of moles of Br_2 used if the product (Y) is treated with excess of $\text{Br}_2 / \text{CCl}_4$?

Q20



Gas Y has been allowed to react with following species in neutral/acidic medium :

- (a) FeCl_3
- (b) CuSO_4
- (c) BaCl_2
- (d) SO_2
- (e) $\text{Cr}_2 \text{O}_7^{2-}$
- (f) $\text{CH}_3 \text{COONa}$
- (g) Hg^{2+}

Then calculate value of $(P - Q)/R$:

P : Number of species which undergoes redox reaction with gas Y.

Q : Number of species with which gas Y undergoes precipitation without going into redox reaction.

R : Number of species with gas Y produce no observable change.

Q21

A gas originally at 1.10 atm and 298 K, underwent a reversible adiabatic expansion to 1.00 atm and 287 K.

What is the molar heat capacity of the gas at constant pressure? (use $\log 287 = 2.45$, $\log 298 = 2.47$ and $\log 110 = 2.04139$)

Q22

How many compounds liberate NH_3 on heating from the following?

- A) $(\text{NH}_4)_2\text{SO}_4$
- B) $(\text{NH}_4)_2\text{CO}_3$
- C) NH_4Cl
- D) NH_4NO_3
- E) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$

Questions with Answer Keys

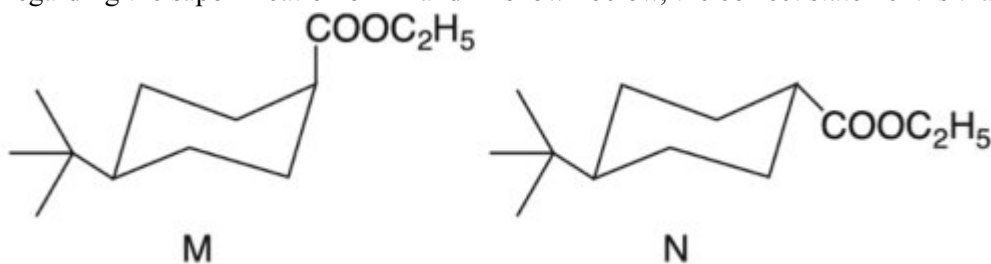
MathonGo

Q23

A symmetrical bridged complex cation made of Co (III), NH_3 molecules and oxygen (in the proper ligand form) is found to have the following composition: Co = 36.875%; NH_3 = 53.125%; O = 10%. The complex cation exists in three ionic forms with cationic charges (A): (n^+) ; (B) : $(n-1)^+$ and (C) : $(n-2)^+$ such that O — O bond length in all of them is found to be more than that in $\text{O}_2 [\text{PtF}_6]$. Calculate the value of 'n' (Nearest Integer). (Atomic weight: Co = 59).

Q24

Regarding the saponification of M and N shown below, the correct statement is that



- (1) M reacts faster than N because the transition state is less crowded for M than for N.
- (2) M reacts slower than N because the transition state is more crowded for M than for N.
- (3) N and M react at the same rate because of formation of tetrahedral intermediate in both cases.
- (4) N reacts slower than M because of its greater thermodynamics stability.

Q25

In paper electrophoresis, amino acids and peptides can be separated by their differential migration in an electric field. To the center of a strip of paper, wet with buffer at pH 6, is applied a mixture of the following three peptides in a single small spot : Gly-Lys, Gly-Asp, and Gly-Ala. A positively charged electrode (anode) is attached to the left side of the paper, and a negatively charged electrode (cathode) to the right side. A voltage is applied across the ends of the paper for a time, after which the peptides have separated into three spots : one near the cathode, one near the anode and one in the center, at the location of the original spot. Which statement is incorrect?

- (1) Gly-Lys migrates to the cathode
- (2) Gly-Asp migrates to the anode
- (3) Gly-Ala does not migrate
- (4) Gly-Lys is a acidic dipeptide

Q26

For a certain orbital

$$\Psi(r) = \frac{1}{16\sqrt{4a_0}}(1 - \sigma)(\sigma^2 - 8\sigma + 12)e^{-\sigma/2}, \sigma = \frac{2r}{na_0} \{a_0 = \text{constant}\}$$

Questions with Answer Keys

MathonGo

The distance of nearest radial node from nucleus is xa_0 while farthest radial node from nucleus is ya_0 .

Find $\left(\frac{y}{x}\right)$.

- (1) 4
(2) 6
(3) 4.5
(4) 3.5

Q27

For given solution in the questions molarity can be treated as molality and solute(s) is/are nonvolatile in nature.

Given that : K_a of $\text{CH}_3\text{COOH} = 10^{-5}$; K_f = Molal depression constant K_a of $\text{HCOOH} = 10^{-4}$; K_b = Molal elevation constant

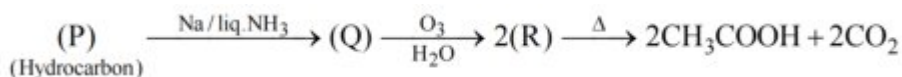
Column - I	Column - II	Column - III
(Colligative properties)	(One of the product formed during electrolysis at Pt - electrode)	(Nature of aqueous solution at 25 C)
(I) $\Delta T_b \simeq 0.3 K_b$	(A) Metal becomes deposited	(P) pH = 8.5 at cathode
(II) $\pi \simeq 0.4RT$	(B) H_2 gas at cathode	(Q) pH = 4.7
(III) $\Delta T_f \simeq 0.2 K_f$	(C) H_2 gas at anode	(R) pH ≈ 7
(IV) $\pi \simeq 0.3RT$	(D) O_2 gas at anode	(S) Weakly basic solution

Which of the following option is correctly matched for 0.1M HCOOK (aq.)

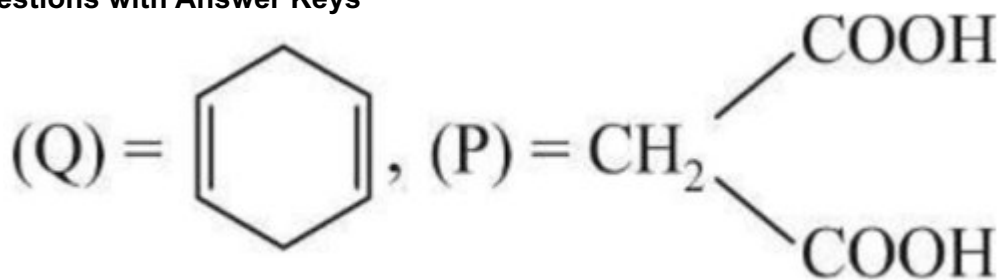
- (1) (I), (B), (S)
(2) (IV), (B), (S)
(3) (III), (D), (Q)
(4) (III), (C), (P)

Q28

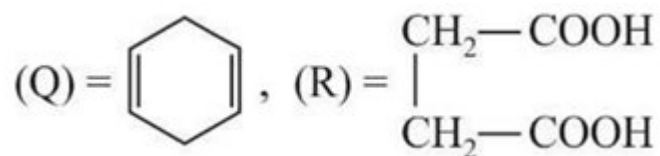
Identify the unknowns in the following reaction sequence.



- (1) (Q) = , (P) = 
(2) (P) = , (Q) = 



(3)



(4)

Q29

In a solid AB having CsCl type structure, if we reverse the pattern of arrangement of AB i.e. A^+ will be where B^- was and B^- is where A^+ was, then which of the following statement is/are true for new structure thus obtained :

- (1) C.N of A^+ will be 8
- (2) C.N of B^- will be 8
- (3) next nearest neighbours of B^- will be 6
- (4) formula of compound will be AB_2

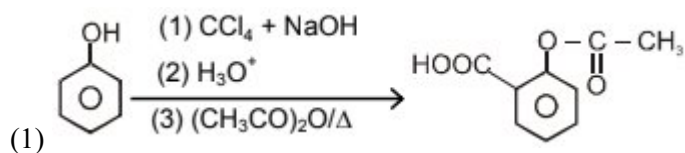
Q30

Saturated solution $Co_2 [Fe(CN)_6]$ is taken in an electrolytic cell of cell constant 0.3 cm^{-1} . If conductivity of saturated solution of $Co_2 [Fe(CN)_6]$ & water used to make solution is $2.2 \times 10^{-6} \text{ S cm}^{-1}$ and $4 \times 10^{-7} \text{ S cm}^{-1}$ respectively, & molar conductance of Co^{+2} & $[Fe(CN)_6]^{-4}$ ions are $80 \text{ S cm}^2 \text{ mol}^{-1}$ & $440 \text{ S cm}^2 \text{ mol}^{-1}$ respectively then

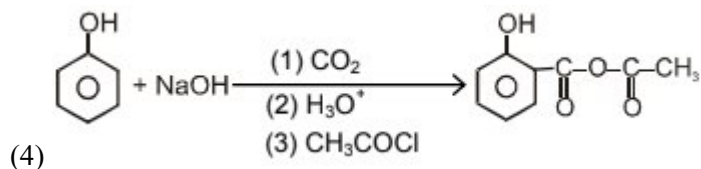
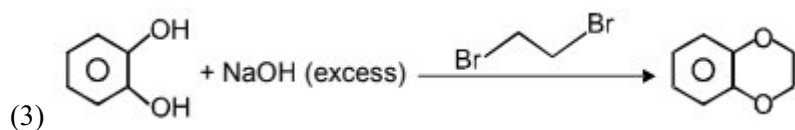
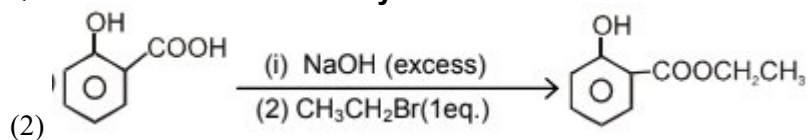
- (1) Solubility of $Co_2 [Fe(CN)_6]$ is $3 \times 10^{-6} \text{ mole/dm}^3$
- (2) K_{SP} of $Co_2 [Fe(CN)_6]$ is $1.08 \times 10^{-16} \text{ mole}^3/\text{dm}^9$
- (3) Ionic mobility of Co^{+2} of this solution increases on increasing potential difference across the terminal of the cell.
- (4) On increasing temperature conductivity of this solution increases.

Q31

In which of the following reaction (s), reactant and products are correctly matched ?



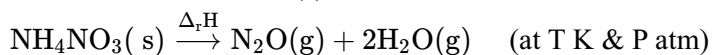
Questions with Answer Keys



Q32

For thermal decomposition of $\text{NH}_4\text{NO}_3(\text{s})$:

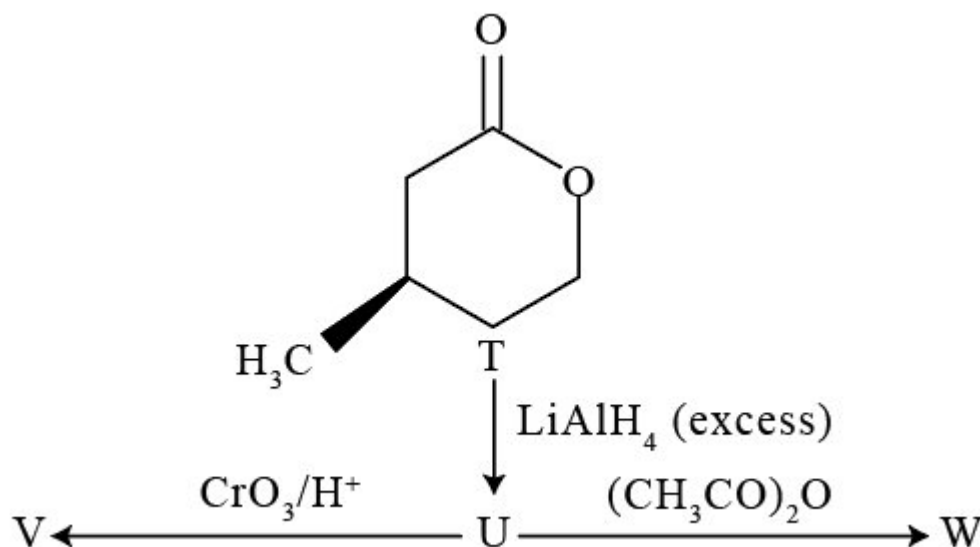
The correct information(s) is/are -



- (1) Heat of reaction is positive
- (2) Entropy change of the reaction must be $\frac{\Delta_r H}{T}$
- (3) Heat of reaction at constant volume is equal to $(\Delta_r H - 3RT)$
- (4) $\Delta_r G$ of this reaction may be negative at high temperature.

Q33

With reference to the scheme given, which of the given statement(s) about T , U , V and W is/are correct?



- (1) T is soluble in hot aqueous NaOH

Questions with Answer Keys

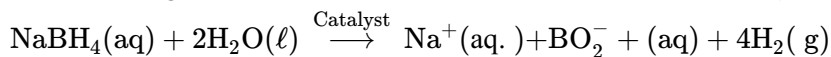
(2) U is optically active

(3) Molecular formula of W is $C_{10}H_{18}O_4$

(4) V gives effervescence on treatment with aqueous $NaHCO_3$

Q34

The following reaction is a slow reaction and hence Ruthenium (molar mass = 101gm) is used as catalyst



Kinetic studies show that reaction is 1st order with respect to catalyst but zero-order with respect to substrate. The rate of hydrogen production per mole of Ruthenium is $92 \text{ mol } H_2 (\text{molRu})^{-1} \text{ min}^{-1}$ at 27°C . Identify CORRECT option(s) $\{R = 0.08 \text{ atm} - \ell/\text{mol} - K\}$

(1) Approximately 0.045 mgRu, must be added to 0.1 dm^3 , $1 \text{ mol dm}^{-3} NaBH_4$ to supply H_2 gas at rate of $0.1 \text{ dm}^3 \text{ min}^{-1}$ at 27°C and 1 atm

(2) Approximately 4.574 mg of Ru, must be added to 0.1 dm^3 , $1 \text{ mol dm}^{-3} NaBH_4$ to supply H_2 gas at rate of $0.1 \text{ dm}^3 \text{ min}^{-1}$ at 27°C and 1 atm

(3) If 0.1 dm^3 , $1 \text{ mol dm}^{-3} NaBH_4$ is used to supply H_2 at at rate of $0.1 \text{ dm}^3 \text{ min}^{-1}$ at 27°C , 1 atm, then upto 96 min hydrogen can be supplied in this way

(4) If 0.1 dm^3 , $1 \text{ mol dm}^{-3} NaBH_4$ is used to supply H_2 at at rate of $0.1 \text{ dm}^3 \text{ min}^{-1}$ at 27°C , 1 atm, then upto 48 min hydrogen can be supplied in this way

Q35

Let $f : R^+ \rightarrow R$ be a differentiable function with $f(1) = 3$ and satisfying

$$\int_1^{xy} f(t)dt = y \int_1^x f(t)dt + x \int_1^y f(t)dt \forall x, y \in R^+, \text{ then the value of } f(e^{37}) \text{ must be}$$

Q36

Let $y = y(x)$ satisfy the differential equation $\left(2xy + x^2y + \frac{y^3}{3}\right)dx + (x^2 + y^2)dy = 0$. If $y(1) = 1$ and the value of $(y(0))^3 = ke (k \in N)$. Find k .

Q37

If equation $4x^4 - ax^3 + bx^2 - cx + 6 = 0$ has four distinct real roots say $x_1 < x_2 < x_3 < x_4$ such that

$$\frac{1}{x_1} + \frac{2}{x_2} + \frac{3}{x_3} + \frac{4}{x_4} = 8, \quad a, b, c \in R^+ \text{ then } \frac{x_4}{x_1} \text{ is equal to}$$

Q38

The area bounded by curves $y = \left[6 + x \left[\frac{1}{x}\right]\right]$, $y^2 - 18x + 18 = 0$ and $6x - 5y - 6 = 0$, (where $[.]$ denotes the greatest integer function) is A , then $A =$

Q39

Questions with Answer Keys

MathonGo

If L.C.M. of $(a, b) = 5^2 \times 3^4$, L.C.M. of $(b, c) = 5^3 \times 3^4$ & L.C.M. of $(c, a) = 5^3 \times 3^4$, then possible ordered triplets of (a, b, c) are k then $\frac{k}{5}$ is equal to

Q40

Let $y = f(x)$ be drawn with $f(0) = 2$ and for each real number a , the tangent to $y = f(x)$ at $(a, f(a))$, has x intercept $(a - 2)$. If $f(x)$ is of the form ke^{px} , then $\left(\frac{k}{p}\right)$ has the value equal to

Q41

Let a three-dimensional vector $\vec{V}$ satisfy the condition, $2\vec{V} + \vec{V} \times (\hat{i} + 2\hat{j}) = 2\hat{i} + \hat{k}$. If $3|\vec{V}| = \sqrt{m}$. Then the value of m is.

- (1) 4
- (2) 5
- (3) 6
- (4) 7

Q42

Consider two polynomial $f(x)$ and $g(x)$ given by $f(x) = \sum_{r=0}^{200} \beta_r x^r$ and $g(x) = \sum_{r=0}^{200} \alpha_r x^r$ such that $\beta_r = 1$ $\forall 100 \leq r \leq 200$ and $f(x+1) = g(x)$. Let $A = \sum_{r=100}^{200} \alpha_r$, then the remainder when A is divided by 15 is equal to

- (1) 11
- (2) 13
- (3) 7
- (4) 1

Q43

Let the circle $(x-1)^2 + (y-2)^2 = 25$ cuts rectangular hyperbola with transverse axis along $y = x$ at 4 points A, B, C and D having coordinates (x_i, y_i) , $i = 1, 2, 3, 4$ such that origin is centre of hyperbola. If

$\ell = x_1 + x_2 + x_3 + x_4$, $m = x_1^2 + x_2^2 + x_3^2 + x_4^2$, $n = y_1^2 + y_2^2 + y_3^2 + y_4^2$ then $\frac{m+n}{\ell} =$

- (1) 100
- (2) 10
- (3) 50
- (4) 20

Questions with Answer Keys

MathonGo

Q44

If $\alpha = \tan^{-1}(1) + \frac{1}{2} \tan^{-1}(2) + \frac{1}{3} \tan^{-1}(3)$, $\beta = \tan^{-1}(1) + 2 \tan^{-1}\left(\frac{1}{2}\right) + 3 \tan^{-1}\left(\frac{1}{3}\right)$ and $|\alpha - \beta| = \frac{p\pi}{q} + \frac{r}{s} \cot^{-1}(3)$, where $p, q, r, s \in N$ and are in their lowest form, then which of the following is INCORRECT?

- (1) $p - r = 0$
- (2) $q = 4s$
- (3) $p + q + r + s = 42$
- (4) $pr = 1 + q$

Q45

Let P_1 denotes the equation of the plane to which the vector $(\hat{i} + \hat{j})$ is normal and which contains the line ' L' ' whose equation is $\vec{r} = (\hat{i} + \hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} - \hat{k})$, P_2 denotes the equation of the plane containing the line ' L ' and a point with position vector $\hat{j}$. Then which of the following is not true :

- (1) Equation of P_1 is $x + y = 2$
- (2) Equation of P_2 is $\vec{r} \cdot (\hat{i} + 2\hat{j} - \hat{k}) = 2$
- (3) The acute angle between P_1 and P_2 is $\cot^{-1}(\sqrt{3})$
- (4) The angle between the plane P_2 and the line ' L' ' is $\tan^{-1} \sqrt{3}$

Q46

Let $A = \{z_1 ; z_1^{16} = 1, z_1 \in C\}$, $B = \{z_2 ; z_2^{72} = 1, z_2 \in C\}$ and $P = \{z_1 z_2 ; z_1 \in A, z_2 \in B\}$ are three sets of complex roots of unity (where C denotes set of complex numbers), then

- (1) $n(A \cap B) = 8$
- (2) $n(A \cap B) = 4$
- (3) $n(P) = 144$
- (4) $n(P) = 72$

Q47

If the series of natural numbers is divided into groups $(1), (2, 3, 4, 5, 6, 7, 8, 9), (10, 11, 12, 13, 14, \dots, 36), \dots$ and so on, then -

- (1) total number of natural numbers in first n groups is $\left(\frac{n(n+1)}{2}\right)^2$

Questions with Answer Keys

MathonGo

(2) first term in n^{th} group is $\frac{1}{4}(n-1)^2n^2 + 1$

(3) last term in n^{th} group is $\frac{1}{4}(n-1)^2n^2 + n^3$

(4) the sum of the numbers in n^{th} group is $\frac{n^3}{4}(n^4 + n^2 + 2)$

Q48

Let $S_n = \lim_{n \rightarrow \infty} \left(\frac{1}{n^6} + \frac{32}{n^6} + \dots + \frac{1}{n} \right)$ and $T_n = \lim_{n \rightarrow \infty} \left(\frac{1}{n^6} + \frac{32}{n^6} + \dots + \frac{(n-1)^5}{n^6} \right)$, then which of the following is/are correct?

(1) $S_n \rightarrow \frac{1}{6}^+$

(2) $(S_n + T_n) < \frac{1}{3}$

(3) $(S_n + T_n) > \frac{1}{3}$

(4) $T_n \rightarrow \frac{1}{6}^-$

Q49

Let matrix $B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ and A is 3 ordered square matrix such that $AB = BA$. If all entries of matrix A are whole numbers whose sum is 6, then-

(1) If there are exactly 3 zero's in matrix A , then $\det(A) = 1$

(2) If trace $A = 6$, then $\det(A) = 8$

(3) A is always an invertible matrix

(4) there are 4 such matrices A .

Q50

If $f(x)$ is a function defined on the set of non negative integers and having co-domain also as set of non negative integers, given that

(i) $x - f(x) = 20 \left[\frac{x}{20} \right] - 80 \left[\frac{f(x)}{80} \right]$ and (ii) $1990 < f(2020) < 2050$. Then possible values of $f(2020)$ can take $[.]$ denote greatest integer function)

(1) 1920

(2) 3960

(3) 2000

(4) 4000

Q51

$x + y = 2$ and $x - y = 2$ are tangents on a parabola at $(1, 1)$ and $(4, 2)$ respectively. Which of the followings is/are correct.

(1) Equation of directrix is $x + 3y = 2$

(2) Equation of axis is $3x - y = 5$

(3) Focus of the parabola is at $\left(\frac{8}{5}, \frac{6}{5}\right)$

(4) Vertex of the parabola is at $\left(\frac{33}{20}, \frac{13}{20}\right)$

ANSWER KEYS

1. (2)	2. (5)	3. (2)	4. (1)	5. (8)	6. (2)
7. (3)	8. (3)	9. (3)	10. (4)	11. (1)	12. (1, 3, 4)
13. (1, 2, 4)	14. (1, 2, 4)	15. (2, 4)	16. (1, 2)	17. (1, 2, 4)	18. (7.5)
19. (0)	20. (0.5)	21. (5)	22. (3)	23. (6)	24. (2)
25. (4)	26. (2)	27. (4)	28. (1)	29. (1, 2, 3)	30. (1, 2, 4)
31. (1, 3)	32. (1, 3, 4)	33. (1, 3, 4)	34. (2, 3)	35. (114)	36. (4)
37. (4)	38. (11)	39. (13.0)	40. (4.0)	41. (3)	42. (4)
43. (3)	44. (3)	45. (4)	46. (1, 3)	47. (1, 2, 3, 4)	48. (1, 3, 4)
49. (1, 2)	50. (1, 3)	51. (1, 3)			